

Environment-sweep training of the DCT-IV operator

DMRG-style closed-form gate updates — no Adam, no learning rate — on the controlled O(2)-twiddle DCT-IV, DIV2K-8q: two inits $\times$ two sweep orders

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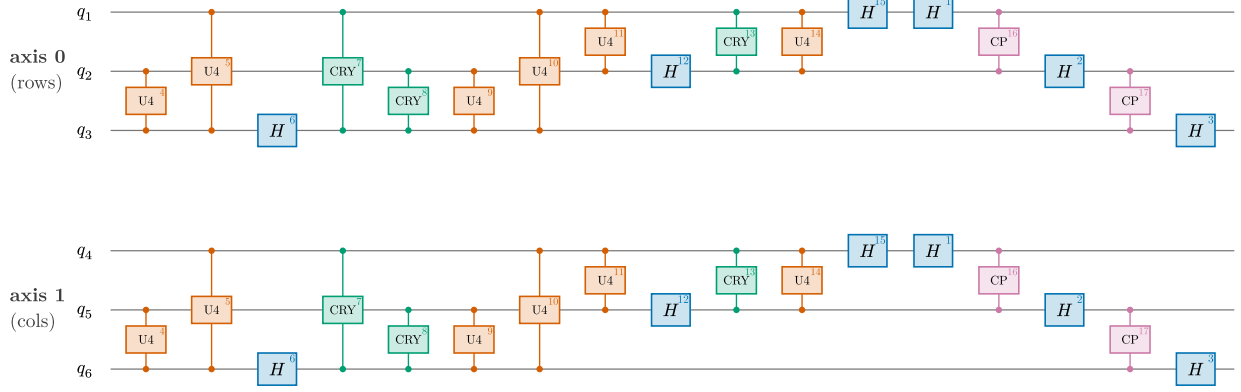
What a sweep update is. For a circuit in which gate G_i enters linearly, the derivative of the loss w.r.t. that gate **is its environment** E_i — the network contracted with everything except G_i . To first order, $\Delta L \propto \langle E_i, \Delta G \rangle$, and the orthogonal matrix minimizing the linearized loss is closed-form: with SVD $E_i = U \Sigma V^\top$, $G_i^* = -UV^\top$ (for the Δ -sign phase gate, $\varphi^* = \pi + \arg E_{11}$). We visit the 214 gate slots one at a time (Gauss-Seidel — each visit re-derives E_i with all previous updates in place), jump to G_i^* behind a backtracking acceptance check ($t \in \{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}\}$ along $\text{polar}((1-t)G_i + tG_i^*)$, accept only on strict loss decrease), and repeat full sweeps until a plateau. The fixed-batch loss is therefore **monotone non-increasing by construction**, with **no learning rate, momentum, or schedule** — the antithesis of the Riemannian Adam used everywhere else in this project, which consumes the same environments as small simultaneous scheduled steps.

Question

Riemannian Adam reaches ≈ 33.2 dB @ $\rho\{=\}.20$ from the exact DCT-IV init and ≈ 29.2 dB from random inits (1008 steps). Does hyperparameter-free coordinate-wise optimization — the classic tensor-network sweep — land in the same place? From both a **random** init and the **exact** DCT-IV init, and does the **visit order** (fwd = circuit emission order, rev = reversed) matter?

Setup

DCT4Basis(8, 8, parametrization: "controlled") (pdf PR #24, commit 5365a5a): 214 gates / 2200 real entries — O(2) singles and CRY twiddle leaves, O(4) mirror gates, Δ -sign phase gates (Figure 1). Loss: top-10% reconstruction MSE ($k = 6554$) on a **fixed** batch of the first 50 images of the canonical seed-42 DIV2K-8q train pool — fixed, so every run is deterministic. Test PSNR on the canonical 50-image test set. Random init: haar-SO per gate (seed 1, shared across orders), the same init family as the seed sweep. Sweeps stop at relative per-sweep improvement < 0.00001 or zero accepted visits; all four runs stopped on this plateau criterion (the exact runs within 18 sweeps, the random runs — extended after an initial 20-sweep pass — within 68), none at a sweep cap.



Superscripts give the **sweep visit order** (basis tensor-slot order): **fwd** visits gates $1 \rightarrow N$, **rev** visits $N \rightarrow 1$.
 $\heartsuit$ The full sweep repeats until the accepted fraction $\rightarrow 0$ — **every** gate is revisited on **every** pass (Gauss-Seidel).

■ **H** Hadamard / branch (1q) ■ **U4** mirror CNOT (2q) ■ **CRY** twiddle R_y (2q) ■ **CP** Δ -sign phase (2q)

Figure 1: The **2-D** DCT-IV(3,3) — two independent DCT-IV(3) blocks, one per image axis (**axis 0** = $q_1..q_3$, rows; **axis 1** = $q_4..q_6$, columns) — a small stand-in for the DIV2K DCT-IV(8,8) (214 gates, axis 1 = $q_9..q_{16}$). The four trainable gate kinds are the labels that appear at the drops in Figure 2: **H** branch Hadamard (1q), **U4** mirror-CNOT (2q), **CRY** single-angle twiddle (2q), **CP** Δ -sign controlled-phase (2q). Superscripts give the **sweep visit order** (the basis' tensor-slot order): a **fwd** sweep visits gates $1 \rightarrow N$, **rev** visits $N \rightarrow 1$, and — unlike the one-gate-per-stage unfreeze study — the whole sweep repeats with **every** gate revisited on **every** pass (Gauss-Seidel) until the accepted fraction falls to ≈ 0 .

Sweep dynamics

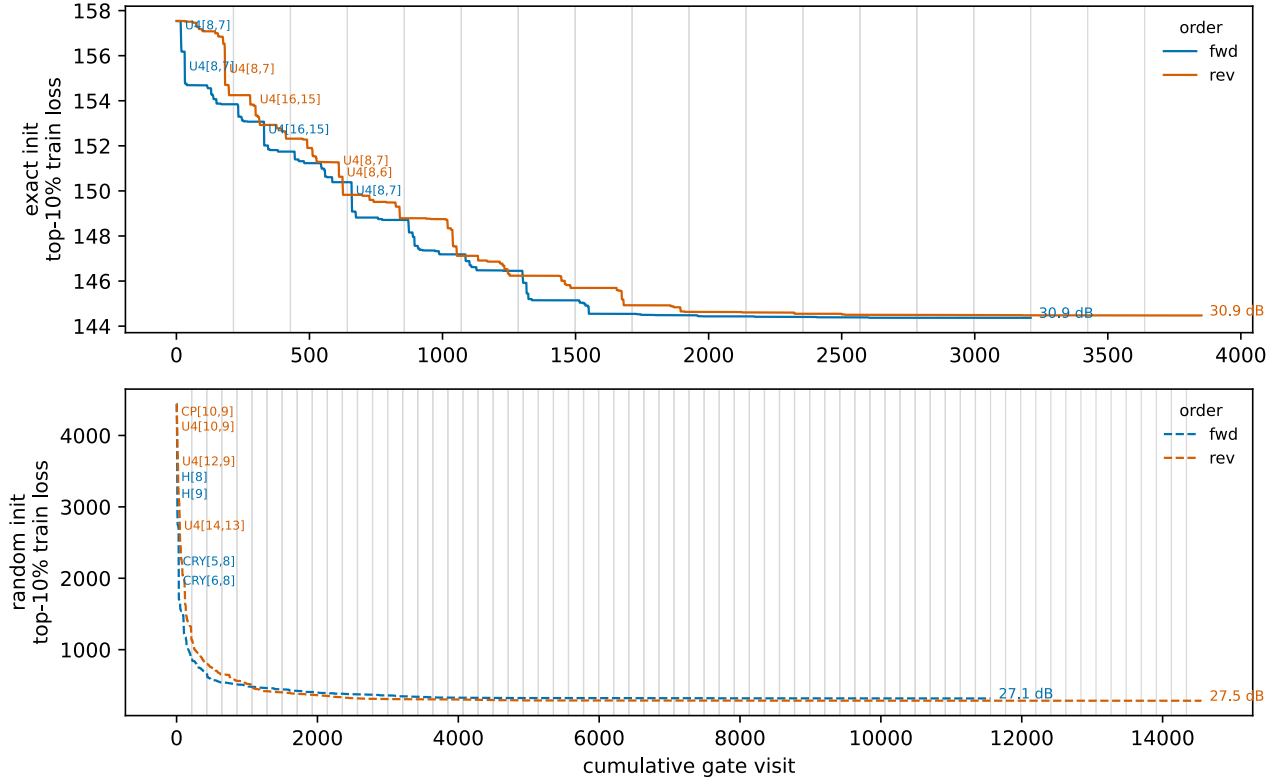


Figure 2: Fixed-batch top-10% loss vs **cumulative gate visit** (top: exact init, bottom: random init; blue = **fwd**, vermilion = **rev**; light rules = sweep boundaries). Labels mark the largest single-visit drops (gate kind [qubits]); endpoints are annotated with test PSNR@ $\rho=\cdot.20$. Every accepted visit decreases the loss by construction.

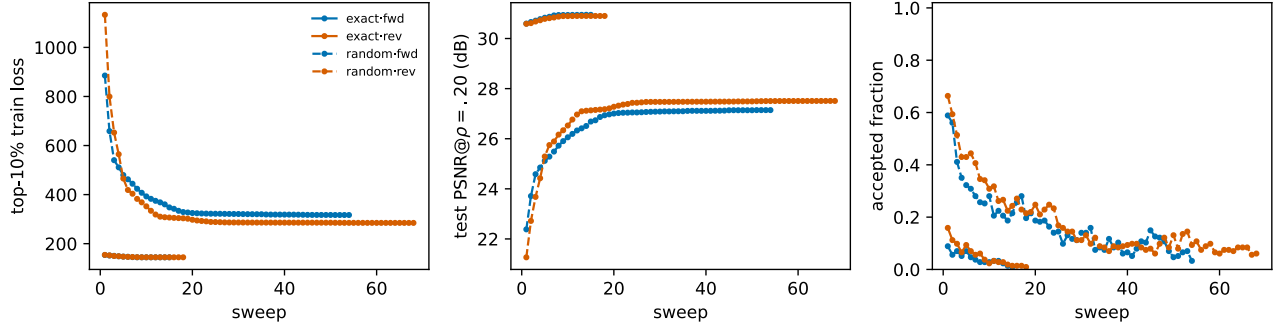


Figure 3: Per-sweep endpoint loss, test PSNR@ $\rho=\cdot.20$, and accepted-visit fraction. The accepted fraction decaying toward zero is the sweep “drying up” — the plateau criterion.

Reconstruction PSNR (DIV2K)

configuration	$\rho=\cdot.05$	$\rho=\cdot.10$	$\rho=\cdot.20$	train MSE	sweeps
sweep · exact/fwd	25.2	27.6	30.9	144.3689	15
sweep · exact/rev	25.1	27.5	30.9	144.4721	18
sweep · random/fwd	22.0	24.1	27.1	316.8243	54
sweep · random/rev	22.5	24.6	27.5	284.662	68
Adam · exact init (1008 steps)	26.5	29.3	33.2	123.1788	—
Adam · random init (mean of 100 seeds) [†]	24.1 ± 0.4	26.1 ± 0.5	29.2 ± 0.5	—	—
classical · DCT-IV (untrained)	24.8	27.2	30.5	—	—

configuration	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.20$	train MSE	sweeps
classical · block-DCT 8×8	26.1	29.4	34.0	—	—
classical · block-FFT 8×8	24.5	27.1	30.8	—	—

train MSE = final fixed-batch top-10% loss (sweep runs; Adam-exact shown on its own 500-image-pool loss — different batch, indicative only). †the random-init Adam reference trained at top-20% (its sweep predates the top-10% convention); the rate-matched probe in the

unfreeze study bounds the effect of the top- k choice at ≤ 0.07 dB.

Reading

From the exact init the sweep barely moves — and only the mirror gates move. Both orders converge (15 and 18 sweeps; the accepted fraction falls to ≈ 0 in the right panel of the convergence figure) to ≈ 30.9 dB @ $\rho\{=\}.20$, just 0.4 dB above the analytic DCT-IV (30.54 dB) and essentially **schedule-independent** — **fwd** and **rev** finish within 0.04 dB of each other and overlap in the dynamics staircase. Every one of the twenty largest single-visit drops is an **O(4) mirror-CNOT (U4)** gate, led by **U4[8,7]**: from the analytic init all the improvement the sweep can find lives in the nominally “fixed-routing” mirror block, while the single-qubit Hadamards and CRY twiddles — already optimal in the exact DCT-IV — never carry a large drop. Yet this converged endpoint sits **2.3 dB below Adam** from the same init (33.2 dB), and still 1.7 dB below at the **matched** top-10% rate $\rho\{=\}.10$: the closed-form local optimum is a genuinely shallower basin than the one scheduled Adam reaches, not a rate artifact.

From the random init the sweep descends fast, then converges — well short of Adam. The loss plunges two orders of magnitude in the first few sweeps — the largest drops now spread across **all** gate kinds (Hadamards, CRY twiddles, mirror CNOTs, and the Δ -sign phase), led by **H[8]**, **CRY[5,8]**, **CRY[6,8]** — and test PSNR climbs from 16.2 to ≈ 27.5 dB @ $\rho\{=\}.20$, where it **flattens**: the accepted fraction decays toward 0 and test PSNR plateaus even as the training loss keeps inching down — mild overfitting of the **fixed** 50-image batch, exactly as in the exact case. Both orders converge: **fwd** in 54 sweeps to 27.14 dB, **rev** in 68 sweeps to 27.51 dB — both ≈ 1.7 –2 dB below Adam from random init (29.2 ± 0.5 dB). This is a **converged** gap, not the stopping artifact it would have been at the original 20-sweep cap (the random curves reach ≈ 0 accepted-fraction in the convergence figure, not the 0.2 plateau of a truncated run). Order matters more here than for the exact init (**rev** 27.51 > **fwd** 27.14 dB).

Same conclusion from both ends. Run to its plateau, the environment sweep **converges below** Riemannian Adam from either init — ≈ 2.3 dB short from exact, ≈ 2 dB from random — even though it is monotone, hyperparameter-free, and robust to initialization and order. Two candidate causes; one is ruled out. The top- k rate does **not** explain the gap: the exact-init deficit is 1.7 dB even at the **matched** $\rho\{=\}.10$. The **fixed 50-image batch**, however, is a genuine confound — on both inits the training loss keeps falling while test PSNR plateaus (the deterministic sweep overfits its batch), whereas the Adam references mini-batched over the full 500-image pool. So part of the deficit is regularization rather than optimizer quality, and cleanly separating the two needs a larger-batch sweep (deferred). What is unambiguous is that greedy, one-gate-at-a-time closed-form solves land in **shallower basins** than simultaneous, momentum-assisted, schedule-annealed updates. For this family the sweep is a reliable way to **polish or sanity-check** an operator. At the headline $\rho\{=\}.20$ every learned DCT-IV here — sweep and Adam alike — still sits below the classical **block-DCT 8×8** (34.0 dB), the strongest transform at that rate; only at the sparsest $\rho\{=\}.05$ does a learned basis edge it (Adam-from-exact 26.5 vs block-DCT 26.1 dB).